ECET 250E-Linear Circuit 1
Problemset 5-Chapter 6- Capacitors and Inductors
Solution

Name: _____ Marks: _____

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1. **6.16** The equivalent capacitance at terminals *a*-*b* in the circuit of **Fig. 6.50** is 30 μF. Calculate the value of *C*.

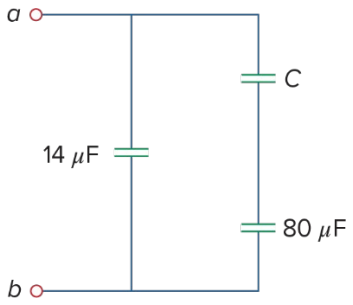


Figure 6.50 For **Prob. 6.16**.

Chapter 6, Solution 16

$$C_{eq} = 14 + \frac{Cx80}{C + 80} = 30 \qquad \boldsymbol{C = 20\ \mu F}$$

2. **6.17** Determine the equivalent capacitance for each of the circuits of **Fig. 6.51**.

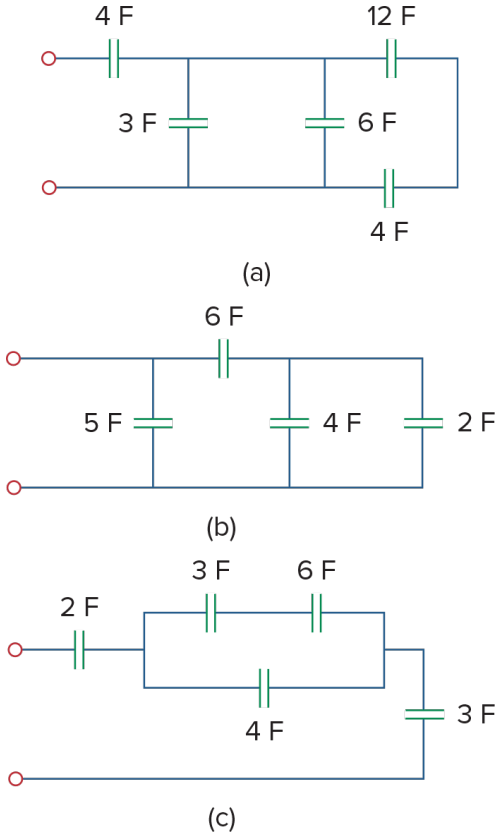


Figure 6.51 For **Prob. 6.17**.

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

Chapter 6, Solution 17.

(a) 4F in series with 12F = 4 x 12/ (16) = 3F

3F in parallel with 6F and 3F = 3+6+3 = 12F

4F in series with 12F = 3F

i.e. C_{eq} = **3F**

(b) $C_{eq} = 5 + [6x (4 + 2)/ (6+4+2)] = 5 + (36/12) = 5 + 3 =$ **8F**

(c) 3F in series with 6F = (3 x 6)/9 = 2F

$$\frac{1}{C_{eq}} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3} = 1$$

C_{eq} = **1F**

3. 6.18 Find Ceq in the circuit of **Fig. 6.52** if all capacitors are 4 μF.

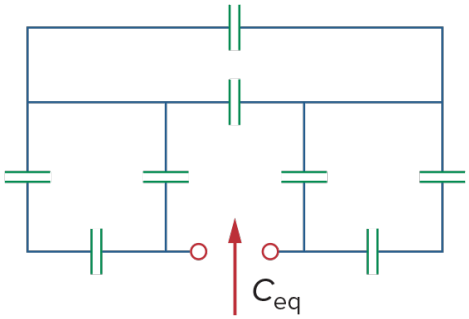


Figure 6.52 For **Prob. 6.18**.

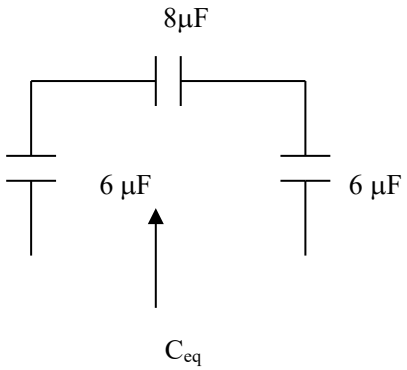
Chapter 6, Solution 18

4 μF in parallel with 4 μF = 8μF

4 μF in series with 4 μF = 2 μF

2 μF in parallel with 4 μF = 6 μF

Hence, the circuit is reduced to that shown below.



$$\frac{1}{C_{eq}} = \frac{1}{6} + \frac{1}{6} + \frac{1}{8} = 0.4583 \longrightarrow C_{eq} = \underline{2.1818\ \mu F}$$

4. **6.26** Three capacitors, C1=5 μF, C2=10 μF, and C3=20 μF, are connected in parallel across a 150-V source. Determine:

1. (a)the total capacitance,
2. (b)the charge on each capacitor,
3. (c)the total energy stored in the parallel combination

(a) $C_{eq} = C_1 + C_2 + C_3 = \mathbf{35\mu F}$

(b) $Q_1 = C_1 v = 5 \times 150\mu C = \mathbf{0.75mC}$

$Q_2 = C_2 v = 10 \times 150\mu C = \mathbf{1.5mC}$

$Q_3 = C_3 v = 20 \times 150 = \mathbf{3mC}$

(c) $w = \frac{1}{2} C_{eq} v^2 = \frac{1}{2} x35x150^2 \mu J = \mathbf{393.8mJ}$

5. **6.32** In the circuit in **Fig. 6.64**, let $i_s = 50e^{-2t}$ mA and $v_1(0)=50$ V, $v_2(0)=20$ V. Determine: (a) $v_1(t)$ and $v_2(t)$, (b) the energy in each capacitor at t=0.5 s.

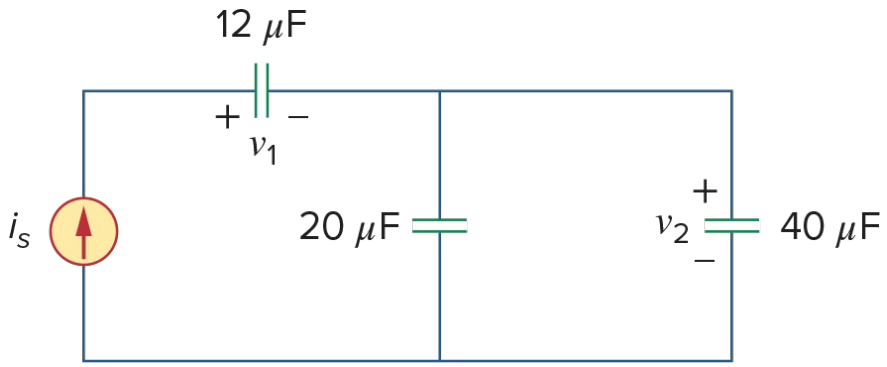


Figure 6.64 For **Prob. 6.32**.

Chapter 6, Solution 32

(a) $C_{eq} = (12x(20||40=60))/(12+60) = 10 \ \mu F$

$$v_1 = \frac{10^{-3}}{12x10^{-6}} \int_0^t 50e^{-2t}dt + v_1(0) = \underline{-2083e^{-2t}\Big|_0^t + 50} = \underline{-2083e^{-2t} + 2133V}$$

$$v_2 = \frac{10^{-3}}{60x10^{-6}} \int_0^t 50e^{-2t}dt + v_2(0) = \underline{-416.7e^{-2t}\Big|_0^t + 20} = \underline{-416.7e^{-2t} + 436.7V}$$

(b) At: t=0.5s,

$$v_1 = -2083e^{-1} + 2133 = 1366.7, \qquad v_2 = -416.7e^{-1} + 436.7 = 283.4$$

$$w_{12\mu F} = \frac{1}{2} x12x10^{-6} x(1366.7)^2 = \underline{11.207 \text{ J}}$$

$$w_{20\mu F} = \frac{1}{2} x20x10^{-6} x(283.4)^2 = \underline{\mathbf{803.2\ mJ}}$$

$$w_{40\mu F} = \frac{1}{2} x40x10^{-6} x(283.4)^2 = \underline{\mathbf{1.6063\ J}}$$

6. **6.40** The current through a 5-mH inductor is shown in **Fig. 6.66**. Determine the voltage across the inductor at t=1, 3, and 5 ms.

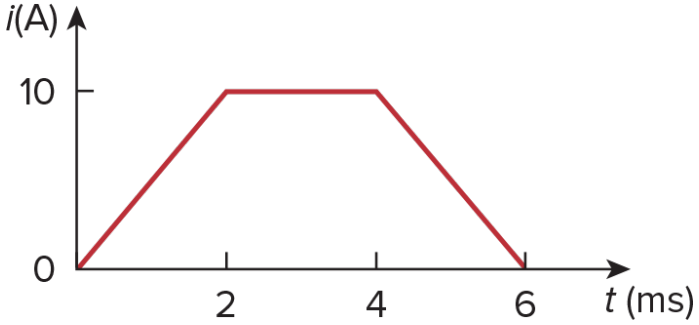


Figure 6.66 For **Prob. 6.40**.

Chapter 6, Solution 40

$$i = \begin{cases} 5t, & 0 < t < 2ms \\ 10, & 2 < t < 4ms \\ 30 - 5t, & 4 < t < 6ms \end{cases}$$

$$v = L\frac{di}{dt} = \frac{5x10^{-3}}{10^{-3}} \begin{cases} 5, & 0 < t < 2ms \\ 0, & 2 < t < 4ms \\ -5, & 4 < t < 6ms \end{cases} = \begin{cases} 25, & 0 < t < 2ms \\ 0, & 2 < t < 4ms \\ -25, & 4 < t < 6ms \end{cases}$$

At t = 1ms, v = **25 V**
At t = 3ms, v = **0 V**
At t = 5ms, v= **–25 V**

7. **6.43** The current in an 80-mH inductor increases from 0 to 60 mA (steady state). How much energy is stored in the inductor?

Chapter 6, Solution 43

$$w = L\int_{-\infty}^{t} i\,di = \frac{1}{2}Li^2(t) - \frac{1}{2}Li^2(-\infty)$$

$$= \frac{1}{2}x80x10^{-3}x\left(60x10^{-3}\right)^2 - 0$$

= **144 μJ**.

8. **6.51** Determine Leq at terminals *a*-*b* of the circuit in **Fig. 6.73**.

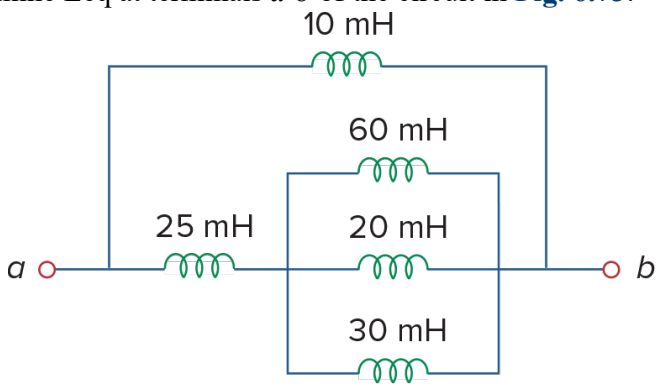


Figure 6.73 For **Prob. 6.51**.

Chapter 6, Solution 51

$$\frac{1}{L} = \frac{1}{60} + \frac{1}{20} + \frac{1}{30} = \frac{1}{10} \qquad L = 10 \text{ mH}$$

$$L_{eq} = 10 \| (25 + 10) = \frac{10x35}{45}$$

= **7.778 mH**

9. **6.53** Find Leq at the terminals of the circuit in **Fig. 6.75**.

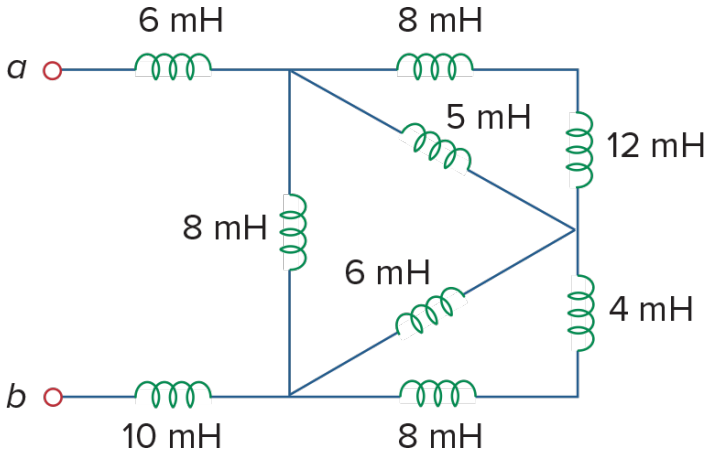


Figure 6.75 For **Prob. 6.53**.

$$L_{eq} = 6 + 10 + 8 \| [5 \| (8+12) + 6 \| (8+4)]$$

$$= 16 + 8 \| (4+4) = 16 + 4$$

L_{eq} = **20 mH**

10. 6.60 In the circuit of **Fig. 6.82**, io(0)=2 A. Determine io(t) and vo(t) for t>0.

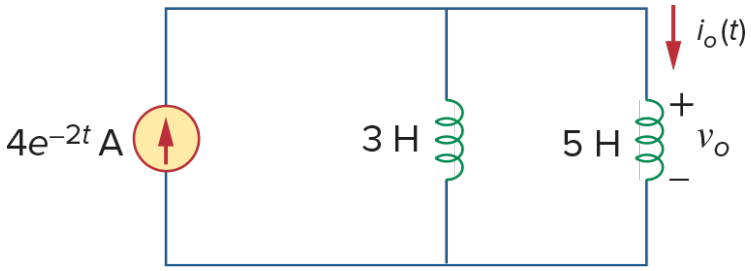


Figure 6.82 For **Prob. 6.60**.

Chapter 6, Solution 60

$$L_{eq} = 3 // 5 = \frac{15}{8}$$

$$v_o = L_{eq}\frac{di}{dt} = \frac{15}{8}\frac{d}{dt}\left(4e^{-2t}\right) = \underline{-15e^{-2t}}$$

$$i_o = \frac{I}{L}\int_0^t v_o(t)dt + i_o(0) = 2 + \frac{1}{5}\int_0^t (-15)e^{-2t}dt = 2 + 1.5e^{-2t}\Big|_0^t$$

i_o = **(0.5 + 1.5e^{-2t}) A**